
ENTANGLEMENT OF ANGULAR MOMENTUM

Entanglement in quantum theory is a consequence of the superposition principle for probability amplitudes. The term relates to a pair of quantum systems whose state cannot be expressed as the product of the single state of each system. The simplest example is the singlet, or zero angular momentum, state of a pair of spin-half particles. This has the form $(1/\sqrt{2})(|+\rangle_1|-\rangle_2 - |-\rangle_1|+\rangle_2)$ where $|+\rangle_i$ is the spin up, and $|-\rangle_i$ the spin down, state for particle i .

Entangled states have remarkable properties. A measurement on one of the entangled particles appears to modify instantaneously the state of its partner, an effect which occurs irrespective of the distance between the particles. The apparent conflict between this result and the predictions of relativity is perhaps the most startling feature of modern quantum theory (8.1).

The bizarre consequences of entanglement were exposed to experimental test through the theoretical contributions of Bell (8.2). He showed that the combination of locality and reality, that is the absence of superluminal influences and the fact that properties exist even if we choose not to measure them, lead to an experimentally testable inequality. Quantum mechanics predicts that this inequality can be violated and so there is a conflict between quantum theory and “local realism”. Aspect and co-workers demonstrated a violation of Bell’s inequality in experiments on the pairs of photons emitted in a $J = 0$ to $J = 1$ to $J = 0$ radiative cascade in calcium (8.3). The polarisations of the two emitted photons are entangled, with a state in the form $(1/\sqrt{2})(|R\rangle_1|R\rangle_2 + |L\rangle_1|L\rangle_2)$. This is a state of entangled spin angular momentum. Because circular polarisation is assigned relative to the direction of propagation, this state of two counter-propagating photons includes two left-handed photons and two right-handed and is thus a state of zero total angular momentum. The evidence of these and subsequent experiments is strongly supportive of quantum mechanics and provides convincing evidence of the existence of entangled states. “Non-local” phenomena associated with entangled polarisation states are an active field of study and play an important role in the emerging field of quantum information (8.4).

Entanglement of orbital angular momentum has recently been demonstrated in spontaneous parametric down-conversion (8.5, **Paper 8.1**). This experiment demonstrated both the conservation of orbital angular momentum in individual down-conversion events at the single-photon level and that the signal and idler photon pair are formed in an entangled superposition of orbital angular momentum states. The form of this entangled state can be derived from the phase matching conditions, or from linear momentum conservation, for the signal and idler photons (8.6, **Paper 8.2**). A simplified

version of this derivation will serve to illustrate the idea. Consider a plane wave pump field propagating in the z -direction. Momentum conservation in the transverse (x - y) plane means that the dependence of the wavefunction for the signal (s) and idler (i) photons must be of the form $\delta(k_{s,x} + k_{i,x})\delta(k_{s,y} + k_{i,y})$. A Fourier transform allows this to be written in the position representation as:

$$\delta(x_s - x_i)\delta(y_s - y_i) = \frac{1}{r_s}\delta(r_s - r_i)\delta(\phi_s - \phi_i) = \frac{1}{2\pi r_s}\delta(r_s - r_i) \sum_{l=-\infty}^{\infty} \exp(il\phi_s) \exp(-il\phi_i),$$

where r and ϕ are the radial and azimuthal coordinates of the position of emission of the photons. The exponential terms are eigenstates of orbital angular momentum and the state, derived on the basis of phase matching, is an entangled state comprising a superposition of states in which the signal and idler orbital angular momenta sum to zero. Naturally, the experimental system is more complicated than this, but the principle that orbital angular momentum entanglement arises from phase matching remains valid. A more complicated picture may occur when extending this idea to more complex crystal geometries (8.7, 8.8, 8.9) beyond the paraxial approximation.

Entanglement is a precious resource for quantum information applications. For this reason it is desirable to have an efficient method for measuring entangled states. A multi-channel analyser capable of determining the orbital angular momentum, l , for a single photon has been built and is described in (8.10, **Paper 8.3**).

While spin angular momentum and polarised light can be characterised by two orthogonal states, orbital angular momentum is higher dimensional (8.11, **Paper 2.4**); multi-dimensional entanglement of orbital angular momentum states might well find interesting applications in quantum information and cryptography.

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Entanglement of the orbital angular momentum states of photons

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Entangled quantum states are not separable, regardless of the spatial separation of their components. This is a manifestation of an aspect of quantum mechanics known as quantum non-locality^{1,2}. An important consequence of this is that the measurement of the state of one particle in a two-particle entangled state defines the state of the second particle instantaneously, whereas neither particle possesses its own well-defined state before the measurement. Experimental realizations of entanglement have hitherto been restricted to two-state quantum systems^{3–6}, involving, for example, the two orthogonal polarization states of photons. Here we demonstrate entanglement involving the spatial modes of the electromagnetic field carrying orbital angular momentum. As these modes can be used to define an infinitely dimensional discrete Hilbert space, this approach provides a practical route to entanglement that involves many orthogonal quantum states, rather than just two. Multi-dimensional entangled states could be of considerable importance in the field of quantum information^{7,8}, enabling, for example, more efficient use of communication channels in quantum cryptography^{9–11}.

Multi-dimensional entanglement is a way—in addition to multi-particle entanglement—to extend the usual two-dimensional two-

particle state. There have been suggestions^{12,13} (and only a proof-of-principle experiment¹⁴) as to how to realize higher-order entanglement via multiport beam splitters. Here we present an experiment in which we used the spatial modes of the electromagnetic field carrying orbital angular momentum to create multi-dimensional entanglement. The advantage of using these modes to create entanglement is that they can be used to define an infinitely dimensional discrete (because of the quantization of angular momentum) Hilbert space.

The experimental realization proceeded in the following two steps. First, we confirmed that spontaneous parametric down-conversion conserves the orbital angular momentum of photons. This was done for pump beams carrying orbital angular momenta of $-\hbar$, 0 and $+\hbar$ per photon, respectively. Second, we showed that the state of the down-converted photons can not be explained by assuming classical correlation—in the sense that the photon pairs produced are just a mixture of the combinations allowed by conservation of angular momentum. We proved that, in contrast, they are a coherent superposition of these combinations, and hence they have to be considered as entangled in their orbital angular momentum. After completion of the experimental work presented here, related theoretical work was brought to our attention^{15,16}.

We will now discuss in order the two steps mentioned above. For paraxial light beams, Laguerre–gaussian (LG) modes (Fig. 1) define a possible set of basis vectors. As predicted¹⁷ and observed¹⁸, LG modes carry an orbital angular momentum for linearly polarized light that is distinct from the intrinsic angular momentum of photons associated with their polarizations. This external angular momentum of the photon states is the reason why they have been suggested for gearing micromachines, and it has been shown that they can be used as optical tweezers^{19–21}.

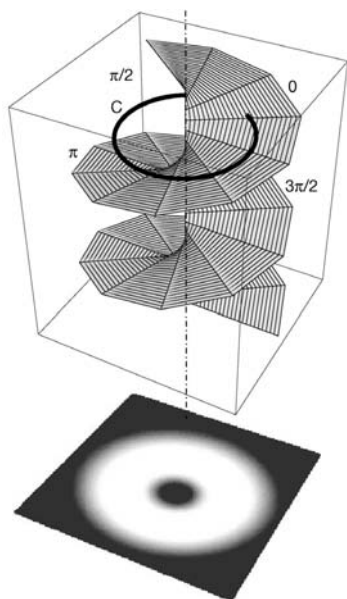


Figure 1 The wave front (top) and the intensity pattern (bottom) of the simplest Laguerre–gaussian (LG_0^1) or ‘doughnut’ mode. The index l is referred to as the winding number, and $(p + 1)$ is the number of radial nodes. Here we only consider cases of $p = 0$. The customary gaussian mode can be viewed as an LG mode with $l = 0$. The handedness of the helical wave fronts of the LG modes is linked to the sign of the index l , and can be chosen by convention. The azimuthal phase term $\exp ikl\phi$ of the LG modes results in helical wave fronts. The phase variation along a closed path C around the beam centre is $2\pi l$. Therefore, in order to fulfil the wave equation, the intensity has to vanish in the centre of the beam.

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To demonstrate the conservation of the orbital angular momentum carried by the LG modes in spontaneous parametric down-conversion, we investigated three different cases—for pump photons possessing orbital angular momenta of $-\hbar$, 0 and $+\hbar$ per photon, respectively. As a pump beam, we used an argon-ion laser (wavelength 351 nm) which we could operate either with a simple gaussian mode profile ($l = 0$) or in the first-order LG modes ($l = \pm 1$) after astigmatic mode conversion (for a description of this technique, see ref. 22). Spontaneous parametric down-conversion was done in a 1.5-mm-thick BBO (β -barium borate) crystal cut for type-I phase matching (that is, both photons carry the same linear polarization). The crystal cut was chosen so as to produce down-converted photons at a wavelength of 702 nm at an angle of 4° off the pump direction.

The mode detection of the down-converted photons was performed for gaussian and LG modes. The gaussian mode ($l = 0$) was identified using mono-mode fibres (Fig. 2) in connection with avalanche detectors. All other modes have a larger spatial extension, and therefore cannot be coupled into the mono-mode fibre. The LG modes ($l \neq 0$) were identified using mode detectors consisting of computer-generated holograms and mono-mode optical fibres (Fig. 2).

Computer-generated holograms have often been exploited for creating LG modes of various orders²³. Our holograms were phase gratings $5 \text{ mm} \times 5 \text{ mm}$ in size, with 20 lines mm^{-1} , which we first recorded on holographic films and bleached afterwards to increase the transmission efficiency (Fig. 2). We made holograms that had one or two dislocations in the centre, and designed them to have their maximum intensity in the first diffraction order. This enabled us to distinguish between LG modes $l = -2, -1, 0, 1, 2$ using all holograms in the first diffraction order—for which they have been blazed. For analysing an LG mode with a negative index, the holograms were rotated by 180° around the axis perpendicular to the grating lines. The total transmission efficiency of all our holograms was about 80%, and they diffracted 18% of the incoming beam into the desired first order. These characteristics were measured at 632 nm wavelength because a laser source at 702 nm was not available.

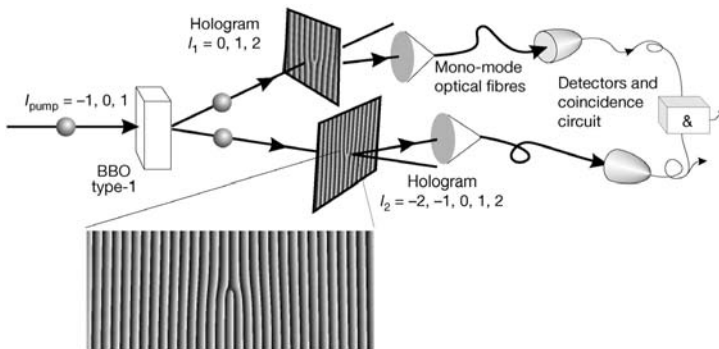


Figure 2 Experimental set-up for single-photon mode detection. After parametric down-conversion, each of the photons enters a mode detector consisting of a computer-generated hologram and a mono-mode optical fibre. By diffraction at the hologram, the incoming mode undergoes a mode transformation in such a way that an LG mode can be transformed into a gaussian mode. As it has a smaller spatial extension than all LG modes, only the gaussian mode can be coupled into the mono-mode fibre. Thus observation of a click projects the mode incident on the fibre coupler into the gaussian mode. The hologram is a phase grating with Δm dislocations in the centre blazed for first-order diffraction. An incoming gaussian laser beam passing through the dislocation of the hologram is diffracted by the grating, and the n th diffraction order becomes an LG mode

The diffraction efficiency is not the only loss that occurs. We also have to account for Fresnel losses at all optical surfaces (95% transmission), imperfect coupling into the optical fibres (70% for a gaussian beam), non-ideal interference filters (75% centre transmission), and the efficiency of the detectors (30%). A conservative estimate of all the losses yields an overall collection efficiency of 2–3%. Comparing the unnormalized ($I_{\text{pump}} = I_1 = I_2 = 0$) coincidence rates of about $2,000 \text{ s}^{-1}$ to the single count rates of about $100,000 \text{ s}^{-1}$ we deduce an efficiency of 2%, in agreement with the above estimation.

The mode analysis was performed in coincidence for all cases where mode filter 1 was prepared for analysing LG modes $l_1 = 0, 1, 2$ and mode filter 2 for those with $l_2 = -2, -1, 0, 1, 2$. For analysing an LG mode with mode index $l = 0$ —that is, a gaussian mode—the dislocation of the hologram was shifted out of the beam path. Thus the beam was sent through the border of the hologram where it acts as a conventional grating without changing the photons' angular momentum. The results are shown in Fig. 3 for different values of orbital angular momenta of the pump beam. Within experimental accuracy, coincidences were only observed in those cases where the sum of the orbital angular momenta of the down-converted photons was equal to the pump beam's orbital angular momentum. However, the absolute count rates of these cases are not equal. This is probably due to unequal emission probabilities of the photons into the different modes in the down-conversion process.

These results confirm conservation of the orbital angular momentum in parametric down-conversion. The signal-to-noise ratios achieved were as high as $V = 0.976 \pm 0.038$ and $V = 0.916 \pm 0.009$ for pump beams with and without orbital angular momentum, respectively. V is defined as $(I_{\text{out}} - I_{\text{in}})/(I_{\text{out}} + I_{\text{in}})$, where I_{in} and I_{out} denote the maximum and the minimum of the coincidences with the dislocation of the hologram respectively in and out of the beam.

It is only by using a coincidence measurement that we could show that the conservation of the orbital angular momentum holds for each single photon pair. In contrast, cumulative detection methods using many photons result in an incoherent pattern²⁴, because

with an index $l = n\Delta m$ and vice versa. Intuitively speaking, the phase dislocation exerts a 'torque' onto the diffracted beam because of the difference of the local grating vectors in the upper and lower parts of the grating. This 'torque' depends on the diffraction order n and on Δm . Consequently the right and left diffraction orders gain different handedness. Reversing this process, a photon with angular momentum $\Delta m\hbar$ before the grating can be detected by the mono-mode fibre detector placed in the first diffraction order. A photon with zero angular momentum (gaussian mode) is detected by diffracting the beam at the border of the hologram far away from the dislocation. All our measurements were performed in coincidence detection between the two down-converted photons.

each beam from parametric down-conversion by itself is an incoherent mixture. Therefore some previous workers²⁴ using these classical detection methods—which are in principle unsuitable at the single photon level—were led to believe that the orbital angular momentum is not conserved in spontaneous parametric down-conversion.

Given this experimental verification of the conservation of orbital angular momentum, entanglement between the two photons produced in the conversion process might be expected. But to explain the conservation of the orbital angular momentum, the photons do not necessarily have to be entangled: it would be sufficient to assume classical correlation. But further experimental results (see below) showed that the two-photon state goes beyond classical correlation, and indeed, we were able to prove the entanglement for photon states with phase singularities.

To confirm entanglement, we have to demonstrate that the two-photon state is not just a mixture but a coherent superposition of product states of the various gaussian and LG modes which obey angular momentum conservation. For simplicity, we restricted ourselves to superpositions of two basis states only. An important distinction between coherent superposition and incoherent mixture of gaussian and LG modes is that the latter possess no phase singularity. This is because adding the spatial intensity distributions of these two modes will yield a finite intensity everywhere in the resulting pattern. In contrast, in a coherent superposition the amplitudes are added, and therefore the phase singularity must remain and is displaced to an eccentric location (Fig. 4). It will appear at that location where the amplitudes of the two modes are equal, with opposite phase. Therefore the radial distance of the singularity from the beam centre is a measure of the amplitude ratio of the gaussian to the LG components, whereas the angular position of the singularity is determined by their relative phase. Intuitively speaking, the position of the dislocation with respect to the beam is equivalent to the orientation of a polarizer.

Superpositions of LG and gaussian modes can be realized experimentally by shifting the dislocation of the hologram out of the centre of the beam by a certain (small) amount. Hence in order to detect a photon having an orbital angular momentum that is a superposition of the gaussian and the LG mode, the hologram was placed in a position such that the dislocation was slightly displaced from the beam centre. In the intensity pattern these modes possess an eccentric singularity (Fig. 4). To demonstrate the entanglement, we therefore shifted one of the holograms and scanned the gaussian mode filter on the other side while recording the coincidences.

The results shown in Fig. 4 verify the correlation in superposition bases of the LG ($l = \pm 2$) and gaussian ($l = 0$) modes. A closer analysis shows that there are two conditions necessary to obtain the

measured curves. First, the shifted hologram has to work as described above, and second, the source must emit an angular-momentum-entangled state. Assume that the source only emits classically correlated but not entangled singularities. Then on the side with the shifted hologram, the various terms of the classical mixture would be projected onto a state with displaced singularity leaving the total state again in a mixture. Respecting the conservation of angular momentum we would then have to sum the probabilities of the various components on the other side, resulting

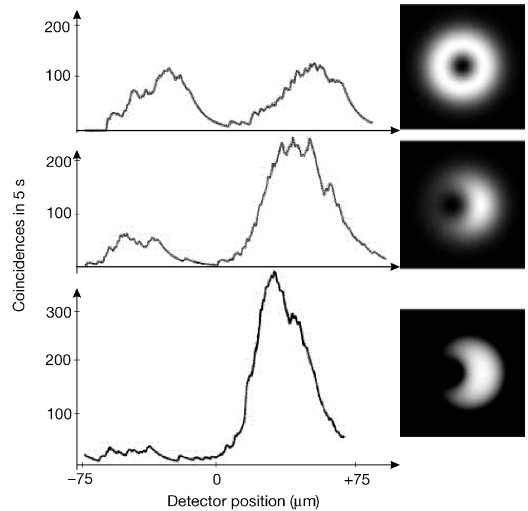


Figure 4 Experimental evidence (left; right, simulation) of entanglement of photon states with phase singularities. The dislocation of the hologram in the beam of photon 1 is shifted out of the beam centre step by step (top, middle, bottom). In these positions, this hologram—together with the mono-mode fibre detector—projects the state of photon 1 into a coherent superposition of LG and gaussian modes. The mode filter for photon 2 with the hologram taken out makes a scan of the second photon's intensity distribution (detector position) in order to identify the location of its singularity with respect to the beam centre. The coincidences show that the second photon is also detected in a superposition of LG and gaussian modes. Classical correlation would yield a coincidence picture which is just a mixture of gaussian and LG modes. In that case, the intensity minimum would remain in the beam centre but would become washed out. In the experiment a hologram with two dislocations in the first diffraction order was used.

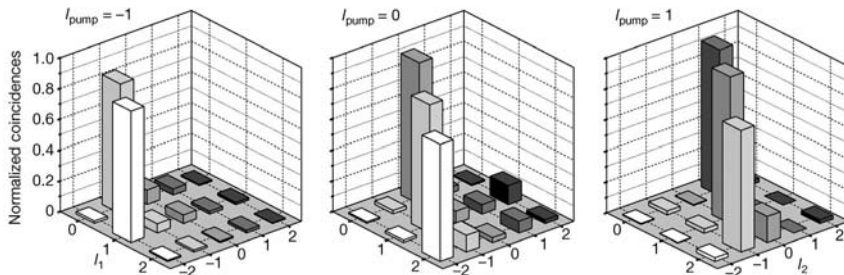


Figure 3 Conservation of orbital angular momentum. Coincidence mode detections for photon 1 and photon 2 in 15 possible combinations of orthogonal states were performed. This was done for a pump beam having an orbital angular momentum of $-\hbar$, 0 and $+\hbar$ per photon, respectively. Coincidences were observed in all cases where the sum of the

orbital angular momenta of the down-converted photons was equal to the pump beam's orbital angular momentum. The coincidence counts for each fixed value of the orbital angular momentum of photon 1 were normalized by the total number of coincidences varying the orbital angular momentum of photon 2.

in a coincidence pattern not containing any intensity zeroes. Such a coincidence pattern would also be observed if a shifted hologram together with a mono-mode detector were not able to analyse for superposition states.

An entangled state represents correctly both the correlation of the eigenmodes and the correlations of their superpositions. Having experimentally confirmed the quantum superposition for $l = 0$ and $l = \pm 2$, it is reasonable to expect the quantum superposition will also occur for the other states. Nevertheless, ultimate confirmation of entanglement will be a Bell inequality experiment generalized to more states²⁵. Such an experiment will be a major experimental challenge, and we are preparing to perform it.

For a pump beam with zero angular momentum, the emitted state must then be represented by

$$\begin{aligned} \psi = & C_{0,0}|0\rangle|0\rangle + C_{1,-1}|1\rangle| -1\rangle + C_{-1,1}| -1\rangle|1\rangle \\ & + C_{2,-2}|2\rangle| -2\rangle + C_{-2,2}| -2\rangle|2\rangle + \dots \end{aligned} \quad (1)$$

as the LG modes form an infinite dimensional basis. Here the numbers in the brackets represent the indices l of the LG modes, and the C_{ij} denote the corresponding probability amplitude for measuring $|i\rangle|j\rangle$. The state (1) is a multi-dimensional entangled state for two photons, which in general will also contain terms with radial mode index $p \neq 0$. It means neither photon in state (1) possesses a well-defined orbital angular momentum after parametric down-conversion. The measurement of one photon defines its orbital angular momentum state, and projects the second one into the corresponding orbital angular momentum state.

It is conceivable that these states could in the future be extended to multi-dimensional multi-particle entanglement. A growing body of theoretical work calls for entanglement of quantum systems of higher dimensions^{7,8}. These states have applications in quantum cryptography with higher alphabets and in quantum teleportation. As such states increase the flux of information, it is conceivable that they could be important for many other applications in quantum communication and in quantum information. The possibility of using these photon states to drive micromachines, and the application of these states as optical tweezers, make them versatile and potentially suitable for future technologies^{19–21}. $\square$

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Two-photon entanglement of orbital angular momentum states

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We investigate the orbital angular momentum correlation of a photon pair created in a spontaneous parametric down-conversion process. We show how the conservation of the orbital angular momentum in this process results from phase matching in the nonlinear crystal.

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Entanglement is one of the most puzzling and powerful properties of quantum theory and has received undiminished attention since the earliest days of quantum mechanics. Entangled states play a crucial role in the investigation of the EPR paradox [1] and in the evaluation of Bell's inequality [2], which distinguishes between local and nonlocal formulations of quantum mechanics. However, two-photon entanglement not only poses fundamental questions to our understanding of quantum theory, but it also plays an important role in applications of quantum mechanics including quantum cryptography, teleportation [3,4], and quantum images [5–7].

Parametric down-conversion has proved a reliable tool for the generation of pairs of entangled photons [8,9]. The resulting spatially separated photons, usually named the “signal” and “idler,” are entangled in their arrival times at the respective detectors [10] and in their transverse positions [6]. They can also be entangled in their polarization states [9,11]. All of these effects have been studied experimentally as well as theoretically. Here we will concentrate on a further entangled property, the orbital angular momentum of the two generated photons [13,14]. The time entanglement arises from the energy conservation in the down-conversion process, which is expressed by the frequency matching condition. Similarly, the entanglement of the transverse position in the far field arises from the momentum conservation expressed by the phase-matching condition.

In this paper, we will show that the orbital angular momentum entanglement, a transverse property of the beam, follows also directly from the phase-matching condition and is related to the conservation of orbital angular momentum. Our approach differs from a recent theoretical paper [12] that dealt with the possible conservation of the combined spin and orbital angular momentum in relation to the susceptibility of the down-conversion crystal. The nonlinear electric susceptibility of the crystal determines the polarization properties of the down-converted photons. In the case of type I down-conversion, the polarization of the signal and the idler photons will be identical, its direction being determined by the polarization of the pump. In the case of type II down-conversion, the two down-converted photons have orthogonal polarizations. In specific setups, this can result in polarization entanglement in each pair of signal and idler photons. For each photon pair, be it generated in type I or type II

parametric down-conversion, phase-matching is fulfilled. Although the nature of the nonlinear susceptibility constrains the polarization of the three interacting waves, it in no way determines the orbital angular momentum of any of the beams. The orbital angular momentum is determined solely by the phase structure of each beam. Our analysis provides the theoretical background to a recent experiment, in which the conservation of the orbital angular momentum in parametric down-conversion has been observed for the first time [13,14].

While the spin angular momentum describes the intrinsic photon spin and corresponds to the optical polarization of light, the orbital angular momentum is associated with the transverse phase front of a light beam. Light with an azimuthal phase dependence $\exp(il\varphi)$ carries a well-defined orbital angular momentum of $l\hbar$ per photon [15]. The associated phase discontinuity produces an intensity null on the beam axis. Such light beams are conveniently described in terms of Laguerre-Gaussian modes, characterized by the mode indices l and p , where $p+1$ gives the number of radial nodes, and $2p+l$ the mode order N .

Laguerre-Gaussian beams can be generated by using holograms which have the form of distorted diffraction gratings with an l -pronged fork dislocation on the beam axis. The first-order diffracted beam then has l intertwined helical wavefronts. Alternatively, such holograms can be used in reverse to detect modes of a particular angular momentum number. In this configuration, if the number of dislocations in the hologram matches the angular momentum number of the incident beam, then the first-order diffracted beam has an on-axis intensity which can be detected. If, however, the number of forks does not match the number of dislocations, then no on-axis intensity results.

The correlation between Laguerre-Gaussian modes was studied in a recent down-conversion experiment [13,14]. For a pump beam with $l_{\text{pump}}=0$, nonvanishing coincidence count rates were measured for the detection of $l_{\text{signal}}=2$ and $l_{\text{idler}}=-2$, whereas no significant coincidence rates were found, for example, for the detection of $l_{\text{signal}}=\pm 2$ and $l_{\text{idler}}=0$. This suggests that the angular momentum is conserved in the down-conversion process, so that $l_{\text{signal}}+l_{\text{idler}}=l_{\text{pump}}=0$.

In the following, we derive the correlation between general transverse modes of the signal and idler photon. We then quantify these correlations for the special case of Laguerre-

Gaussian modes and show how these are related to the conservation of the orbital angular momentum.

Here we are concerned with the transverse-mode correlations between the signal and idler beams. Hence we describe the modes of the pump, signal, and idler by the normalized transverse mode functions $\Phi_{0,1,2}(\mathbf{x})$, where $\mathbf{x}$ is a two-dimensional vector in the plane perpendicular to the propagation direction of the light; we work in the paraxial limit. Quantities and operators concerning the pump, signal, and idler are indicated by the subscripts 0, 1, and 2, respectively. For simplicity, we suppress the explicit beam propagation and assume frequency matching to be fulfilled.

We first determine the state of the light generated by a parametric down-conversion process [16,17]. In the Schrödinger picture, the two photons in the signal and the idler mode are created by applying the operator $\hat{a}^\dagger(\mathbf{k}_1)\hat{a}^\dagger(\mathbf{k}_2)$ to the initial vacuum state $|0\rangle$, where $\mathbf{k}_{1,2}$ is the transverse component of the signal and idler wave vector, respectively. Similarly, $\mathbf{k}_0$ denotes the transverse component of the pump beam. We assume polarization states that are in agreement with the condition imposed by the nonlinear susceptibility of the crystal. For photons in such polarization states, phase-matching between the down-converted photons and the pump is satisfied. It can be described by a sinc function $\Pi_{j=x,y} \text{sinc}[(\mathbf{k}_0 - \mathbf{k}_1 - \mathbf{k}_2)_j L_j / 2]$, where L_j is the length of the crystal in the directions transverse to the beam propagation. For simplicity, we assume that the transverse dimension of the crystal is sufficiently large so that the phase-matching condition can be described by a two-dimensional delta function $\delta^{(2)}(\mathbf{k}_0 - \mathbf{k}_1 - \mathbf{k}_2)$ and that all other modes are damped out [16,17]. Moreover, the fact that $\mathbf{k}_0 - \mathbf{k}_1 - \mathbf{k}_2 = 0$ is a good approximation for fields carrying orbital angular momentum has already been experimentally verified in the frequency up-conversion of Laguerre-Gaussian modes [18]. While the δ function constrains the sum of the transverse signal and idler wave vectors, their absolute difference $|\mathbf{k}_1 - \mathbf{k}_2|$ cannot be arbitrarily large either. This is immediately apparent within the paraxial limit, the regime in which most down-conversion experiments are operating. More generally one can argue that large values of $\mathbf{k}_1 - \mathbf{k}_2$ require large values of $\mathbf{k}_1$ and/or $\mathbf{k}_2$ and hence of the signal and idler frequencies ω_1 and ω_2 . Increasing $|\mathbf{k}_1 - \mathbf{k}_2|$ to an arbitrarily large value, therefore, would lead to a violation of the frequency-matching condition $\omega_0 = \omega_1 + \omega_2$ associated with energy conservation. Clearly, energy conservation requires $|\mathbf{k}_1 - \mathbf{k}_2| \leq 2\pi/\lambda$, where λ is the pump wavelength. We include this constraint in our analysis by means of the purely geometrical function $\Delta(\mathbf{k}_1 - \mathbf{k}_2)$, normalized such that $\int d\mathbf{k} |\Delta(\mathbf{k})|^2 = 1$. This function will be zero for large values of its argument so as to satisfy the requirement of energy conservation. In practice, other constraints including the sizes of apertures used and crystal geometry will prescribe the precise form of $\Delta(\mathbf{k}_1 - \mathbf{k}_2)$. We note that this function will result only in a scaling factor for the single and coincidence count rates. It cancels in the normalized count rates and plays no part in the derivation of orbital angular momentum conservation.

The two-photon wave function of the signal and idler then takes the form [19]

$$|\Psi\rangle = \int d\mathbf{k}_0 \int d\mathbf{k}_1 \int d\mathbf{k}_2 \Phi_0(\mathbf{k}_0) \hat{a}_2^\dagger(\mathbf{k}_2) \hat{a}_1^\dagger(\mathbf{k}_1) \times \Delta(\mathbf{k}_1 - \mathbf{k}_2) \delta^{(2)}(\mathbf{k}_0 - \mathbf{k}_1 - \mathbf{k}_2) |0\rangle. \quad (1)$$

Here

$$\Phi(\mathbf{k}_{0,1,2}) = \frac{1}{2\pi} \int d\mathbf{x} \Phi_{0,1,2}(\mathbf{x}_{0,1,2}) \exp(i\mathbf{k}_{0,1,2} \cdot \mathbf{x}_{0,1,2}) \quad (2)$$

denote the normalized mode functions in Fourier space. Similarly, the Fourier transformation of the creation operators is given by

$$\hat{a}_{1,2}^\dagger(\mathbf{k}_{1,2}) = \frac{1}{2\pi} \int d\mathbf{x} \hat{a}_{1,2}^\dagger(\mathbf{x}_{1,2}) \exp(-i\mathbf{k}_{1,2} \cdot \mathbf{x}_{1,2}). \quad (3)$$

Using these equations, we can write the two-photon wave function (1) in the position representation:

$$|\Psi\rangle = \int d\mathbf{x}_1 \int d\mathbf{x}_2 \Phi_0\left(\frac{\mathbf{x}_1 + \mathbf{x}_2}{2}\right) \times \Delta(\mathbf{x}_1 - \mathbf{x}_2) \hat{a}_1^\dagger(\mathbf{x}_1) \hat{a}_2^\dagger(\mathbf{x}_2) |0\rangle. \quad (4)$$

This is the normalized wave function of the combined system of the signal and idler generated by parametric down-conversion. It contains all necessary information about the outcome of single or coincidence measurements. The count rates of such measurements are proportional to the probabilities of detecting a photon of the signal or idler in the desired mode. In order to calculate these probabilities, we need to find the overlap of the two-photon state with the normalized one-photon state of the signal or idler,

$$|\Psi_{1,2}\rangle = \int d\mathbf{x}_{1,2} \Phi_{1,2}(\mathbf{x}_{1,2}) \hat{a}_{1,2}^\dagger(\mathbf{x}_{1,2}) |0\rangle, \quad (5)$$

associated with detecting a photon in the mode $\Phi_{1,2}$.

The coincidence probability for finding one photon in the signal mode Φ_1 and one photon in the idler mode Φ_2 is then given by

$$P(\Phi_1, \Phi_2) = |\langle \Psi_2, \Psi_1 | \Psi \rangle|^2 = \left| \int d\mathbf{x}_1 \int d\mathbf{x}_2 \Phi_1^*(\mathbf{x}_1) \Phi_2^*(\mathbf{x}_2) \Phi_0 \times \left(\frac{\mathbf{x}_1 + \mathbf{x}_2}{2} \right) \Delta(\mathbf{x}_1 - \mathbf{x}_2) \right|^2. \quad (6)$$

If we average this expression over all possible signal modes, then we obtain the probability for finding a photon in the idler mode,

$$\begin{aligned}
P(\Phi_2) &= |\langle \Psi_2 | \Psi \rangle|^2 \\
&= \int d\mathbf{x}_1 \int d\mathbf{x}_2 \int d\mathbf{x}'_2 \Phi_2(\mathbf{x}'_2) \Phi_2^*(\mathbf{x}_2) \Phi_0^* \left(\frac{\mathbf{x}_1 + \mathbf{x}'_2}{2} \right) \\
&\quad \times \Phi_0 \left(\frac{\mathbf{x}_1 + \mathbf{x}_2}{2} \right) \Delta^*(\mathbf{x}_1 - \mathbf{x}'_2) \Delta(\mathbf{x}_1 - \mathbf{x}_2), \quad (7)
\end{aligned}$$

and a similar expression describes the probability of finding a single photon in the signal mode.

We recall that $\Delta(\mathbf{k})$ is a very broad function that cuts out modes with large transverse wave vectors. The mode functions will therefore vary very little over the region where $\Delta(\mathbf{x}_1 - \mathbf{x}_2)$, the Fourier transform of $\Delta(\mathbf{k})$, does not vanish. Under this assumption, Eq. (6) and (7) can be written as

$$P(\Phi_1, \Phi_2) = \left| \int d\mathbf{y} \Delta(\mathbf{y}) \right|^2 \left| \int d\mathbf{x} \Phi_1^*(\mathbf{x}) \Phi_2^*(\mathbf{x}) \Phi_0(\mathbf{x}) \right|^2, \quad (8)$$

$$P(\Phi_2) = \left| \int d\mathbf{y} \Delta(\mathbf{y}) \right|^2 \int d\mathbf{x} |\Phi_2^*(\mathbf{x}) \Phi_0(\mathbf{x})|^2. \quad (9)$$

The factor $|\int d\mathbf{y} \Delta(\mathbf{y})|^2$ will in general be very small and limit the count rates. This is because the down-converted photons are emitted into a wide spatial range.

The normalized coincidence probability, however, is independent of $\Delta(\mathbf{y})$:

$$\begin{aligned}
P(\Phi_1, \Phi_2)^N &= \frac{P(\Phi_1, \Phi_2)}{\sqrt{P(\Phi_1)P(\Phi_2)}} \\
&= \frac{\left| \int d\mathbf{x} \Phi_1^*(\mathbf{x}) \Phi_2^*(\mathbf{x}) \Phi_0(\mathbf{x}) \right|^2}{\sqrt{\int d\mathbf{x} |\Phi_2^*(\mathbf{x}) \Phi_0(\mathbf{x})|^2} \sqrt{\int d\mathbf{x} |\Phi_1^*(\mathbf{x}) \Phi_0(\mathbf{x})|^2}}. \quad (10)
\end{aligned}$$

This probability can take values between 0 and 1. It becomes 1 if signal and idler are perfectly correlated so that the detection of the signal in mode Φ_1 implies that the idler is in mode Φ_2 , and it vanishes if signal and idler are anticorrelated.

We now need to connect the idler mode function Φ_0 with the mode functions of the down-converted signal and idler $\Phi_{1,2}$. More precisely, we want to determine the idler mode for a given pump mode and detected signal mode. The state of the idler photon collapses into $|\Psi_2\rangle = \langle \Psi_1 | \Psi \rangle$ if the signal photon was found to be in state $|\Psi_1\rangle$. We can calculate this state from Eqs. (4) and (5),

$$|\Psi_2\rangle = \int d\mathbf{y} \Delta(\mathbf{y}) \int d\mathbf{x}_2 \Phi_1^*(\mathbf{x}_2) \Phi_0(\mathbf{x}_2) \hat{a}^\dagger(\mathbf{x}_2) |0\rangle, \quad (11)$$

where we have evaluated the function $\Delta(\mathbf{y})$ in the same way as previously. In comparison with Eq. (5), we find that the

mode function of the idler can be expressed as the product of the mode functions of the signal and the pump,

$$\Phi_2 = \left(\int d\mathbf{y} \Delta(\mathbf{y}) \right) \Phi_0 \Phi_1^*, \quad (12)$$

where the integral enforces the normalization of the idler mode. This may be interpreted “backwards” or retroactively [20]: The measured mode function Φ_2 is “reflected” at the $\chi^{(2)}$ crystal, where it interacts with the pump mode Φ_0 and is modified into the idler mode function.

So far our considerations have been valid for any transverse mode function, including Hermite or Laguerre-Gaussian modes. In the following, we will concentrate on Laguerre-Gaussian modes and specifically investigate the correlation of the orbital angular momentum between the signal and the idler. The normalized Laguerre-Gaussian modes in polar coordinates are given by

$$\begin{aligned}
\Phi_{p,l}(r, \varphi) &= \sqrt{\frac{2p!}{\pi(l+p)!}} \sqrt{\frac{1}{w} \left(\frac{r\sqrt{2}}{w} \right)^{|l|}} \\
&\quad \times L_p^{|l|} \left(\frac{2r^2}{w^2} \right) e^{-r^2/w^2} e^{-il\varphi}, \quad (13)
\end{aligned}$$

where the z -dependent phase was omitted and $L_p^{|l|}$ denotes the associated Laguerre polynomial,

$$L_p^l = \sum_{m=0}^p (-1)^m C_{p-m}^{p+l} r^m / m!. \quad (14)$$

Laguerre-Gaussian modes with an integer orbital angular momentum of $\hbar l$ per photon are orthonormal solutions of the paraxial wave equation. We note that these modes can be defined in the same way for fractional values of l . These fractional modes are still normalized, but form an overcomplete set. They can be written as sums of stable Laguerre-Gaussian modes with different integer l 's. As each of these modes has a different Gouy phase, the resulting beams with fractional l 's are unstable and do not maintain their amplitude distribution upon propagation. It has been suggested, however, that such beams can be prepared or detected by use of holograms or phaseplates [21] or by using nonlinear optical devices, such as the optical parametric oscillator [22].

By inserting the spatial mode functions $\Phi_{p_{0,1,2}, l_{0,1,2}}$ in Eq. (12) and comparing the phase factors, we find that the orbital angular momentum of the idler is given by $l_2 = l_0 - l_1$. This signifies that the orbital angular momentum must be conserved,

$$l_0 = l_1 + l_2. \quad (15)$$

We want to stress that we have derived this equation solely from the phase-matching condition for Laguerre-Gaussian modes or other modes with the same azimuthal phase dependence. Our derivation is independent of the nature of the nonlinear susceptibility of the down-conversion crystal.

One example of the conservation of the orbital angular momentum can be seen in Fig. 1, where the mode profile of the idler mode function is shown as the product of the complex pump mode and the signal mode according to Eq. (12). Our derivation of Eq. (15) is valid for all modes that can be

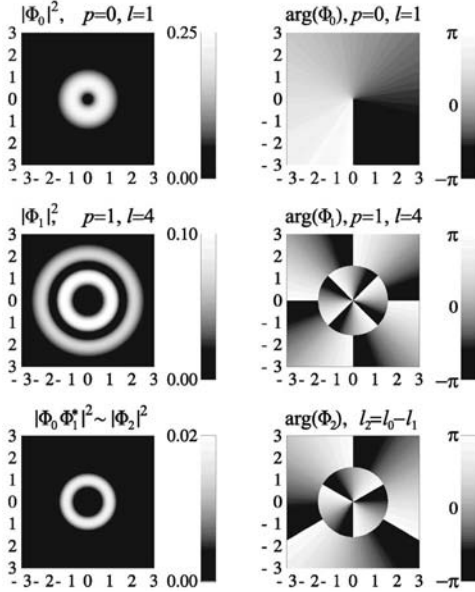


FIG. 1. Intensity $|\Phi|^2$ and phase $\arg(\Phi)$ of the pump, signal, and idler mode. If the pump and signal beams are prepared/measured in the indicated L-G modes, the idler collapses into a mode with an amplitude proportional to $\Phi_0\Phi_1^*$. The phase structure of the idler corresponds to $l_2 = l_0 - l_1$ and an orbital angular momentum of $3\hbar$.

expressed in the form (13). We note that the conservation of orbital angular momentum can be fulfilled not only for integer quantities $l_{0,1,2}$ but also for continuous values of the orbital angular momentum numbers.

Equation (15) is in agreement with the experimental observation of orbital angular momentum conservation reported in [13,14]. To our knowledge, this is the only relevant experiment at the single-photon level. Classical signal and idler beams consist of many photon pairs, for each of which the conservation $l_0 = l_1 + l_2$ is satisfied. Both signal and idler beams will then exhibit modes with mixtures of various l 's rather than pure Laguerre-Gaussian modes [23]. These mixed signal and idler modes will still be correlated. However, unless we can observe each pair of photons, their correlation will be lost as there is no coherence between photons generated from different pump photons. This makes it very difficult to demonstrate the validity of Eq. (15) in the classical regime. In contrast to this, the orbital angular momentum of each photon created by a frequency-doubling process is uniquely determined as $l_{2\omega} = 2l_\omega$, where l_ω denotes the orbital angular momentum of the incoming photons. Consequently, the conservation of orbital angular momentum can be observed in classical up-conversion experiments [18].

The l conservation entangles the angular momentum modes of the signal and idler for a given pump mode. We set the waist of the idler equal to the waist of the signal, $w_1 = w_2 = w$, and denote the fraction of the signal and idler waist to the pump waist by $W = w/w_0$. By inserting the form

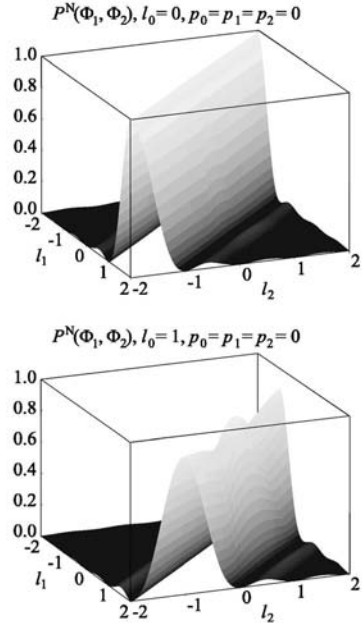


FIG. 2. The normalized coincidence probability $P^N(\Phi_1, \Phi_2)$ for the orbital angular momenta of the signal and idler photons for $l_0 = 0$ and 1. The ratio of idler and signal waists to that of the pump is $W = 0.4$.

of Laguerre-Gaussian modes (13) into the normalized coincidence probability (10), we find

$$P^N(\Phi_1, \Phi_2) = \text{sinc}^2[(l_1 + l_2 - l_0)\pi] \frac{|R_{12}|^2}{\sqrt{R_2 R_1}}. \quad (16)$$

Here we have denoted

$$R_{12} = \int_0^\infty dr r^{(|l_1|+|l_2|+|l_0|)/2} e^{-r(1+W^2/2)} \times L_{p_1}^{|l_1|}(r) L_{p_1}^{|l_2|}(r) L_{p_0}^{|l_0|}(r W^2), \quad (17)$$

$$R_1 = \frac{(p_1 + |l_1|)!}{p_1!} \times \int_0^\infty dr r^{|l_2|+|l_0|} e^{-r(1+W^2)} [L_{p_1}^{|l_1|}(r)]^2 [L_{p_0}^{|l_0|}(r W^2)]^2, \quad (18)$$

and similarly for R_2 . Figure 2 shows the normalized coincidence probability as a function of the continuous orbital angular momenta of the signal and idler photon.

The sinc function in Eq. (15) becomes maximal if the orbital angular momentum is conserved and vanishes if $(l_0 - l_1 - l_2)$ is a nonzero integer. It is therefore most likely to detect a pair of signal and idler modes that conserve the pump angular momentum and it is impossible to find combinations of signal and idler modes that violate angular mo-

mentum conservation by an integer. Secondary maxima of the sinc function near half-integer values of $(l_0 - l_1 - l_2)$ seem to imply that there is a violation of the conservation of orbital angular momentum. This is not the case, as the beams of fractional l may be expressed in terms of modes with integer l , some of which satisfy the conservation requirement. This is due to the fact that the fractional l modes are overcomplete. While this structure is determined by the phase dependence of the L-G modes, we find additional dips along the diagonals of maximum coincidence at $l_1 + l_2 = l_0$. These are related to the radial mode profile in Eq. (16) and again arise from the finite overlap of the fractional modes (12). There is no equivalent to this in polarization correlations. However, it should be stressed that this additional structure arises from the radial structure of the mode. If this radial structure is ignored, as in a detection process that measures only l , then this detailed structure disappears.

Compared with the spin angular momentum, orbital angular momentum offers a far richer structure. The spin angular momentum can only take values between -1 and 1 for right and left circularly polarized light, respectively. The orbital angular momentum is in principle unlimited, and there exists an infinite number of Laguerre-Gaussian modes. While the spin angular momentum modes are defined in a two-dimensional Hilbert space, the orbital angular momentum modes occupy different Hilbert spaces depending on the mode order $N = l + 2p$ with dimensionality $N + 1$ [24].

Orbital angular momentum states with mode order $N = 1$ are exactly analogous to spin angular momentum states. In each case they can be depicted on the Poincaré sphere [25]. Any two diametrically opposed points on the Poincaré sphere correspond to orthogonal states. For the polarization states these are left and right circular polarization, and horizontal and vertical linear polarization. For the orbital angular momentum states these are Laguerre-Gaussian modes with positive and negative phase, vertical and horizontal Hermite-Gaussian modes, or superpositions of such modes. It is well known [26] that maximum violation of Bell's inequality occurs between any two measurements made for polarization states separated by 45° on the Poincaré sphere. For orbital

angular momentum, states, we would therefore anticipate a maximum violation of Bell's inequality to be obtained for measurements of superpositions of Laguerre-Gaussian and Hermite-Gaussian modes of mode order 1 separated by 45° degree on the Poincaré sphere. For larger values of orbital angular momentum, this analogy breaks down due to the higher number of possible states. The quantum correlations, however, persist and offer the prospect of novel demonstrations and applications of entanglement.

Another approach has been proposed by Mair *et al.* [14], whereby a hologram displaced from its on-axis position no longer produces or detects a pure Laguerre-Gaussian mode but a complicated superposition of modes with differing indices. This situation can be modeled by a numerical evaluation of Eq. (10), which we will report elsewhere [27]. In general, the coincidence rate is maximized for hologram offsets, which result in mode combinations with high contributions by mode pairs which satisfy the conservation of orbital angular momentum (15), namely $l_0 = l_1 + l_2$.

In this paper, we have considered the correlation between Laguerre-Gaussian modes of the signal and idler with varying amounts of orbital angular momentum. The finite mode overlap between modes differing by a fractional number of l is to some extent reminiscent of the finite overlap between nonorthogonal polarization states, as they are used for testing Bell's inequality. This suggests the possibility of observing a violation of Bell's inequality for measurements of angular momentum states differing by fractional values of l . Such measurements may be possible with the use of holograms that impose a fractional change of the orbital angular momentum by means of a fractional change of the phase step.

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Measuring the Orbital Angular Momentum of a Single Photon

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We propose an interferometric method for measuring the orbital angular momentum of single photons. We demonstrate its viability by sorting four different orbital angular momentum states, and are thus able to encode two bits of information on a single photon. This new approach has implications for entanglement experiments, quantum cryptography and high density information transfer.

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It is well known that photons can carry both spin and orbital angular momentum (OAM) [1,2]. The spin is associated with polarization and the OAM with the azimuthal phase of the complex electric field. Each photon of a beam with an azimuthal phase dependence of the form $\exp(il\phi)$, for example, carries an OAM of $l\hbar$. The polarization of a single photon is described by a state in a two-dimensional space. For this reason, photon polarization provides a useful physical realization of a single qubit and has been widely employed in demonstrations of quantum key distribution [3,4]. As recently pointed out [5], the infinite number of orthogonal states of OAM places no limit on the number of bits that can be carried by a single photon. Moreover, the ability to create states with different OAM and superpositions of these allows the realization of quNits, that is quantum states in an N -dimensional space, with single photons. The problem addressed in this Letter is the realization of a multichannel device for determining the OAM of a single photon. Such a device will allow us to take advantage of the increase in information capacity associated with orbital rather than spin angular momentum.

Previous work has shown that the OAM of a laser beam containing many photons in the same mode can be measured. For example, interfering an $\exp(il\phi)$ beam with its mirror image produces an interferogram with $2l$ radial spokes [6,7] [Fig. 1(a)]. Although this technique can discriminate between an arbitrarily large number of states, the state of one single photon cannot be measured as many photons are required to form the full interference pattern.

For individual photons, computer-generated holograms [8,9], when used in combination with a pinhole in a setup similar to the one shown in Fig. 1(b), can determine the particular OAM state [10]. Holograms are often used in the generation of $\exp(il\phi)$ beams, where the fork dislocation in the hologram introduces helical phase fronts in the diffracted beam. When operated in reverse, the hologram “flattens” the helical phase fronts. The beam, now with planar phase fronts, can be focused through a pinhole and detected. However, this technique allows photons to be tested only for one particular state. A more complex computer-generated hologram was demonstrated that can detect several different l states but with an efficiency that

cannot exceed the reciprocal of the number of different l values [11]. This low efficiency means that this method is not likely to be useful for quantum information applications. In a different approach, the difference in Gouy phase was used within an interferometer to distinguish between modes of two different orders (the mode order is dependent on l) [12]. Although working at the single-photon level such a scheme sorts between only two states.

In this Letter we describe an interferometric technique that can distinguish individual photons in arbitrarily many OAM states with a theoretical efficiency of 100%. We demonstrate this principle with a device that simultaneously sorts four different OAM states, corresponding to

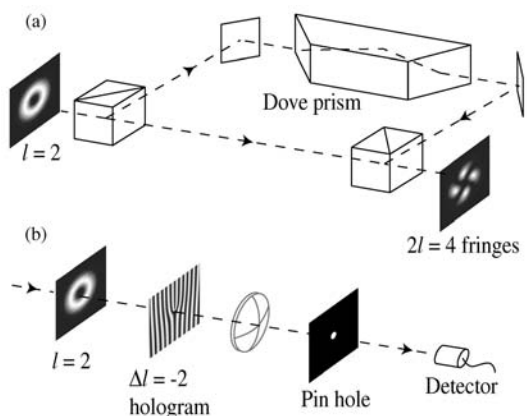


FIG. 1. Previous work on measuring the OAM of light. (a) A Mach-Zehnder interferometer with a Dove prism inserted into one arm interferes the incoming light beam with its own mirror image. In the case of light with l intertwined helical phase fronts, the interference pattern has $2l$ radial fringes. This setup is capable of distinguishing between an arbitrary number of states, but forming the required fringe pattern needs many photons. (b) A hologram can be used to “flatten” the phase fronts of light modes with specific values of l , which makes it possible to focus these modes (but no others) through a pinhole, behind which they can be detected. While this latter setup works with individual photons, it can only test for one particular OAM state.

two bits of information that can be transferred with a single photon.

Our device relies on the $\exp(il\phi)$ form of the transverse modes. On rotation of the beam through an angle α , this phase dependence becomes $\exp[i l(\phi + \alpha)]$. This corresponds to a phase shift of $\Delta\psi = l\alpha$ [13], which is a manifestation of a geometrical phase [14]. For particular combinations of l and α , the rotated beam may be either in or out of phase with respect to the original. For example, when $\alpha = \pi$, a beam with even l is in phase with the original but a beam with odd l rotated by the same angle is out of phase by π (Fig. 2). If such a rotation is incorporated into the arms of a two-beam interferometer, then the phase shift between the two arms becomes l dependent. It follows that for different angles of rotation, constructive and destructive interference occurs for different values of l .

This concept can be realized in the form of a Mach-Zehnder interferometer with a Dove prism inserted into each arm (Fig. 3). A Dove prism flips the transverse cross section of any transmitted beam [15]. Two Dove prisms, rotated with respect to each other through an angle $\alpha/2$,

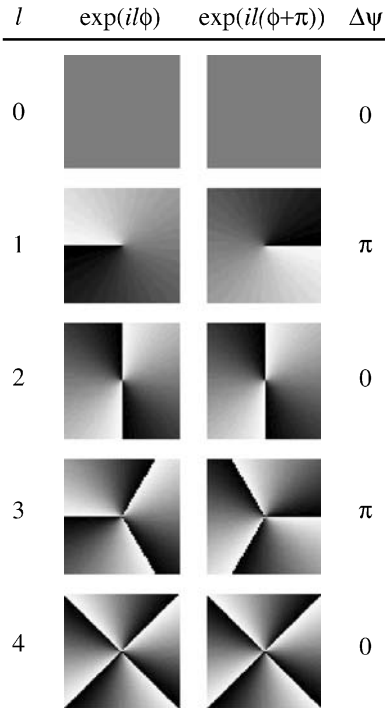


FIG. 2. Gray-scale representations of phase profiles of non-rotated and rotated beams with an $\exp(il\phi)$ phase structure. After a rotation through π , a beam with even l is unchanged, while one with odd l is out of phase by π with the nonrotated beam. Interfering an l beam with a rotated copy of itself therefore results in constructive interference for even l and destructive interference for odd l .

rotate a passing beam through an angle α . In the example shown in Fig. 3, $\alpha/2 = \pi/2$ and hence the relative phase difference between the two arms of the interferometer is $\Delta\psi = l\pi$. By correctly adjusting the path length of the interferometer we can ensure that photons with even l appear in port A1 and photons with odd l appear in port B1. If the input state is a mixture of even and odd l components, then these components are “sorted” into an even channel A1 and an odd channel B1.

Our principle can be extended further to enable us to test for an arbitrarily large number of OAM states. This is achieved by cascading additional Mach-Zehnder interferometers with different rotation angles (Fig. 4). (Note that the scheme outlined in Ref. [12] could be extended in an analogous fashion.) The first interferometer, stage 1, sorts photons with even and odd values of l into ports A1 and B1, respectively. Photons with even l are then passed into the second stage where they are sorted further. The angle between the Dove prisms of the second stage is $\alpha/2 = \pi/4$ corresponding to $\Delta\psi = l\pi/2$. Therefore, modes with $l = 4n$, where n is an integer, go into port A2 and beams with a phase term of $l = 4n + 2$ go into port B2. Unfortunately, there is no rotation angle that allows us to unambiguously sort odd- l photons in the same way. We solve this problem by placing a hologram in front of one interferometer of the second stage so that we can increase the azimuthal phase of the odd- l photons by 1, thereby making their l values even. An additional interferometer with $\alpha = \pi/2$ will now separate the original odd- l photons in the same way as the even- l photons were sorted. Figure 4 outlines the first three sorting stages, which allow discrimination between eight different values of l . By adding further stages, this procedure can be extended to allow an arbitrarily large number of OAM states to be distinguished. It should be noted that, in the absence of holograms, a scheme similar to that illustrated in Fig. 4 can be constructed to sort beams where l takes on the values of 0 or 2^n , where n is an integer.

To demonstrate the viability of our proposed mechanism, three Mach-Zehnder interferometers were built to

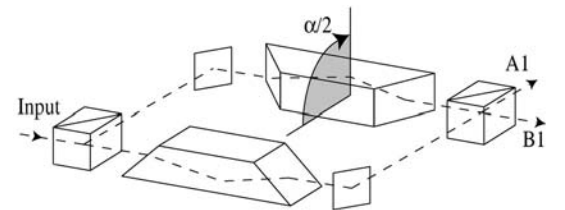


FIG. 3. First stage of our OAM sorter. A Mach-Zehnder interferometer with a Dove prism placed in each arm. The beams in the two arms are rotated with respect to each other through an angle α , where $\alpha/2$ is the relative angle between the dove prisms. In the example shown, $\alpha/2 = \pi/2$, this device sorts photons with even values of l into Port A1 and those with odd values of l into Port B1.

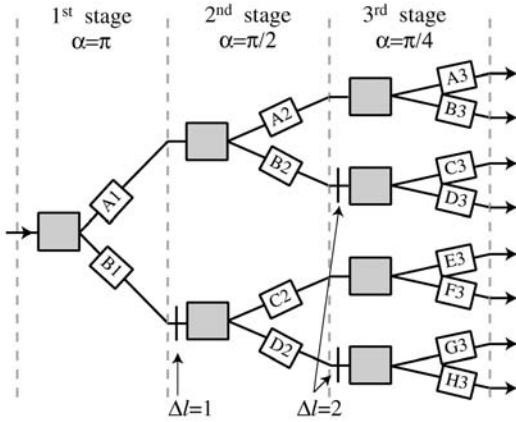


FIG. 4. First three stages of a general sorting scheme. The gray boxes each represent an interferometer of the form shown in Fig. 3 with different angles between the Dove prisms. The first stage introduces a phase shift of $\alpha = \pi$ and so sorts multiples of 2: even l s into Port A1 and odd l s into Port B1. The odd- l photons then pass through a $\Delta l = 1$ hologram so that they become even- l photons. The second stage introduces a phase shift of $\alpha = \pi/2$, so it sorts even- l photons into even and odd multiples of 2. The $\Delta l = 2$ hologram is required before the photons are sorted further in the third stage.

form the first two stages of the general OAM sorter outlined in Fig. 4. The light source used in this experiment was a helium-neon laser with a power of <1 mW. An intracavity cross wire introduced rectangular symmetry to the laser cavity and forced the laser to oscillate in high-order Hermite-Gaussian ($HG_{m,n}$) modes. Such modes are characterized by the indices m and n which correspond to zeros of intensity in the electric field in the x and y directions, respectively. The Hermite-Gaussian modes were then converted to Laguerre-Gaussian modes by passing them through a $\pi/2$ mode converter based on cylindrical lenses [16]. The resulting Laguerre-Gaussian modes have an $\exp(il\phi)$ phase structure and corresponding OAM of $l\hbar$ per photon. This conversion of Hermite-Gaussian ($HG_{m,n}$) beams gives Laguerre-Gaussian (LG_p^l) beams characterized by $l = |m - n|$ and $p = \min(m, n)$. Adjustments to the intracavity cross wire allowed us to generate $HG_{m,0}$ modes with $m = 0, 1, 2, \dots$, which in turn gave rise to LG_0^l beams with $l = 0, 1, 2, \dots$. The interferometers had an arm length of approximately 30 cm and were built from standard optical components. The $\Delta l = 1$ hologram was manufactured using standard photographic techniques [17]. Note that such a hologram increases the l value of any $\exp(il\phi)$ mode by 1 [18]. The four ports were directed onto a screen so a camera could take an image of the output.

Figure 5 shows the output from the two-stage sorting process. As can be seen, we succeeded in sorting modes from $l = 0$ to $l = 4$ into different ports. The $l = 4$ mode appears in the same port as the $l = 0$ beam, as one would

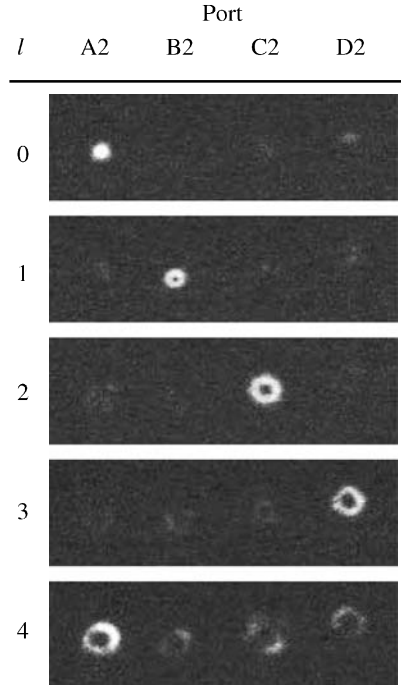


FIG. 5. Experimental results of a 2-stage sorting scheme. The four different output ports correspond to $\exp(il\phi)$ modes with the values $l(\bmod 4) = 0, 1, 2, 3$, respectively.

expect. In this first experiment, the overall efficiency of the OAM sorter was limited by the poor optical efficiency of the particular hologram used to approximately 10%.

To demonstrate that our device works at the single-photon level, a further experiment was carried out at intensities so low that on average less than one photon was present in each interferometer at any one time. This was achieved by inserting neutral-density filters to attenuate the power of the laser beam to <0.3 nW. This experiment used a 1-stage interferometer. The output ports of the interferometer were directed into a camera that averaged over a number of frames. As anticipated, this interferometer still sorts between odd and even l s with an efficiency limited only by the quality of the optical components (Fig. 6). Although the device proposed in this Letter can sort individual photons according to their OAM, we did not detect photons individually. We plan to do this in the future with the use of a single photon source and single photon detectors.

Our OAM sorter is the analog of the polarizing beam splitter in that it selects the optical path on the basis of OAM, one path for each of the distinguishable states. In this way, our sorter can be used to generate entanglement between the optical path and OAM in the same way that a polarizing beam splitter can create entanglement between the optical path and polarization [19]. This will make it

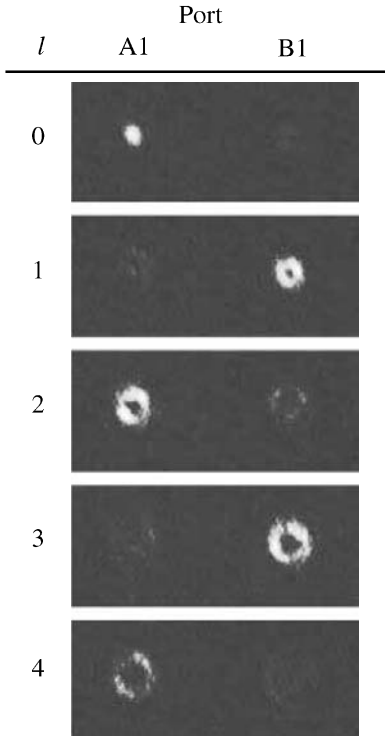


FIG. 6. Experimental results of a 1-stage (even-odd) sorting scheme at the single-photon level. Neutral density filters were used to reduce the number of photons so that the intensity corresponded to one photon in the interferometer at one time.

useful in generating highly entangled states and extending the optical realization of quantum logic elements to OAM quNits [20].

We have demonstrated experimentally that a single photon in an OAM eigenstate can be measured in any one of a number of different orthogonal states corresponding to different values of l (the OAM in units of $\hbar$). Our approach is in principle 100% efficient, limited only by the efficiency of the components.

The ability to measure a single photon to be in any one of an arbitrarily large number of orthogonal states has a number of potential implications for quantum information processing. The efficient measurement of the OAM of a single photon allows us access to a larger state space than that associated with optical polarization. This provides the possibility of a greater density of information transfer along with the generation and analysis of entanglement involving large numbers of states [10]. The implications of this work for entanglement based applications such as superdense coding [21], teleportation [22], and quantum computation [23] remain to be explored.

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